

Business Data Analyzer

Desktop business analytics application for sales, inventory, profitability, and data quality analysis.

Overview

Business Data Analyzer helps you:

- Analyze sales and inventory data
- Validate data quality before reporting
- Detect join and mapping issues
- Calculate profitability and margins
- Export formatted Excel reports
- Explore trends and business insights

Built with:

- Python
 - Streamlit
 - Pandas
-

Features

Sales Analysis

- Revenue analysis
 - Sales trends
 - Product performance
 - Customer performance
 - Profitability analysis
 - Margin analysis
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Inventory Analysis

- Stock visibility
- Low stock detection

- Overstock identification
 - Reorder monitoring
 - Supplier analysis
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Smart Data Preparation

- Automatic column mapping
 - Alias-aware matching
 - Manual mapping override
 - Validation reporting
 - Join diagnostics
 - Preflight risk scoring
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Reporting

- Excel export
 - Validation diagnostics
 - Cleaned datasets
 - Analysis summaries
-

System Requirements

- Windows 10 or later
 - Python 3.10+
 - Internet connection required only for Python package installation on first launch
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Installation

1. Install Python

Download Python:

urlPython Downloads<https://www.python.org/downloads/>

During installation:

☒ Tick:

Add Python to PATH

2. Extract the Application

Extract the downloaded ZIP file to any folder.

Example:

C:\BusinessDataAnalyzer

3. Launch the App

Double-click:

run_app.bat

On first launch:

- A virtual environment will be created
- Required packages will be installed
- Your browser will open automatically

This may take several minutes on first run.

Using the App

Option 1 — Load Demo Data

Use built-in demo datasets to explore features quickly.

Available demos:

- Sales Demo
 - Inventory Demo
 - Full Demo (Sales + Customer + Product Masters)
-

Option 2 — Upload Your Own Data

Supported formats:

- CSV
- XLSX

- XLS

Recommended datasets:

- Sales exports
 - Inventory exports
 - Customer master data
 - Product master data
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Profitability Analysis Requirements

Profitability analysis requires:

- Quantity
 - Unit Price
 - Either:
 - Unit Cost in the sales file
 - OR Product Master with Base Cost
-

Example Data Fields

Sales Dataset

Business Field	Example Column Names
Date	Order Date
Product	Product Name
Quantity	Qty Sold
Unit Price	Selling Price
Unit Cost	Unit Cost
Customer	Client Name

Inventory Dataset

Business Field	Example Column Names
Product	Product Name
SKU	SKU Code
Stock On Hand	Current Stock
Reorder Level	Minimum Stock

Business Field	Example Column Names
Supplier	Supplier Name

Troubleshooting

Browser does not open

Open manually:

`http://localhost:8501`

Python not found

Reinstall Python and ensure:

Add Python to PATH

was enabled during installation.

First launch is slow

The application installs required packages on first run.

Subsequent launches will be much faster.

Folder Structure

```
BusinessDataAnalyzer/  
├── app.py  
├── run_app.bat  
├── requirements.txt  
├── analytics/  
├── config/  
├── data_cleaning/  
├── reporting/  
├── utils/  
├── demo_data/  
└── screenshots/
```

Version

v1.0

Included:

- Sales analysis
 - Inventory analysis
 - Profitability analysis
 - Smart column mapping
 - Validation reporting
 - Excel export
 - Demo mode
-

Planned Features

- Saved mapping profiles
 - Forecasting
 - Customer segmentation
 - Product segmentation
 - Run history
 - Expanded test coverage
-

Support

If you encounter issues during review/testing:

- Include screenshots where possible
 - Include sample files if relevant
 - Include error messages shown in the console or browser
-

License

Internal review build.

Screenshots

Dashboard Overview

Controls

i

Help

- Not sure what to upload? Start with sales or inventory data.
- Missing columns? Use column mapping.
- Profit not showing? Add cost data.

Primary dataset type

auto

Preflight mode

strict

Preview rows

20

☒ Enable manual column mapping

Upload one main dataset, then optionally enrich it with master data.

>

i

How to use this tool

Business Data Analyzer

Clean messy spreadsheets and turn them into business-ready analysis reports.

Upload Your Data

Start with your main dataset. You can optionally enrich it with master data.

Main dataset (sales, inventory, or general)

Upload

200MB per file • CSV, XLSX, XLS

Try Demo Data

Inventory

Load Demo

Optional customer master

Upload

200MB per file • CSV, XLSX, XLS

Optional product master

Upload

200MB per file • CSV, XLSX, XLS

Upload a dataset or try demo data to begin.

Column Mapping

Column Mapping: Sales

Map uploaded columns to the business fields your analysis expects.

☒ ⚡ Auto-map high confidence columns

Required fields for sales: date, product, quantity

> date → date ● High (1.00) ⚡ auto-mapped

> product → product ● High (1.00) ⚡ auto-mapped

> sku_code → sku ● High (1.00) ⚡ auto-mapped

> client_name → customer ● High (1.00) ⚡ auto-mapped

> client_id → customer_id ● High (1.00) ⚡ auto-mapped

> quantity → quantity ● High (1.00) ⚡ auto-mapped

> unit_price → unit_price ● High (1.00) ⚡ auto-mapped

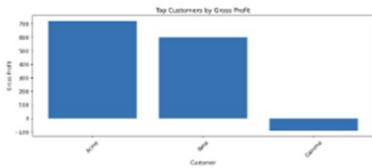
> cost → unit_cost ● High (1.00) ⚡ auto-mapped

Required fields are mapped.

Profitability Analysis

Profitability is calculated using sales and cost data. Negative margins may indicate pricing issues or incorrect cost data.

Profitability



- > Profit by product data
- > Profit by customer data
- > Product profitability segmentation
- > Customer profitability segmentation

Lowest-profit products

	product	revenue	gross_profit	units
0	Widget C	150	-90	3
1	Widget B	2400	600	12
2	Widget A	1800	720	18

Lowest-profit customers

	customer	revenue	gross_profit	units
0	Gamma	150	-90	3
1	Beta	2400	600	12
2	Acme	1800	720	18

Inventory Analysis

Validation Report

1 rows have negative stock levels. | column: stock_on_hand | rows: 1

Sample rows for negative_stock

	snapshot_date	warehouse	product	sku	category	sub_category	supplier_id	supplier	lead_time_days	stock_on_hand	reorder_level
15	2025-12-31	Pretoria Hub	Widget Delta	SKU004	Hardware	Tools	SUP002	BluePeak Components	21	-1159	10

2 rows are at or below reorder level. | rows: 2

Sample rows for low_stock_positions

	snapshot_date	warehouse	product	sku	category	sub_category	supplier_id	supplier	lead_time_days	stock_on_hand	reorder_level
15	2025-12-31	Pretoria Hub	Widget Delta	SKU004	Hardware	Tools	SUP002	BluePeak Components	21	-1159	10
27	2025/12/31	Pretoria Hub	Widget Nova	SKU007	Accessories	Batteries	SUP004	MetroSource Wholesale	10	40	10

